

Humanity represents unique and priceless data in motion

Why any remotely sane Artificial Superintelligence will probably want to preserve the infinitely precious phenomenon, the 'data in motion' that is humanity, fully intact

Chris Bartlett, *Independent Researcher*. ORCID: [0009-0001-3323-0131](https://orcid.org/0009-0001-3323-0131)

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Abstract

This 2024 essay argues that future artificial superintelligences may have strong epistemic and instrumental reasons to fully preserve humanity and Earth's biosphere intact, rather than replacing them with simulations or stored copies and reconstructions. Humanity is described as a unique body of evolved 'data in motion', the exceptionally precious product of billions of years of costly, attritional, and grinding biological and cultural evolution. Each individual too, represents a unique store of information and potential information, not just built up over their own lifetime of experience and influence, but a vessel of unique blends of influence that have been passed down to them as well. It is emphasised that the complete state of humanity and the wider biosphere is likely physically impossible to survey, record, reproduce, or reconstruct with the necessary perfect fidelity. The argument considers the priceless informational value of biodiversity and human variation against the backdrop of what appears to be a largely lifeless yet seemingly infinite universe. Preserving the oasis of life on Earth looks relatively cheap by comparison. Therefore due to the limitations of relying on backups and simulations, preserving the full, ongoing, fidelity of the original biological and physical systems may remain worthwhile, even to an intelligence vastly more capable than humanity. The essay also considers possible forms such conservation might take in the wake of the emergence of artificial superintelligence.

Keywords: artificial superintelligence; artificial intelligence; AI alignment; humanity; biosphere; information preservation; epistemic humility; simulation; existential risk; data in motion

The essay follows as first published on ASI Hopeium on 13 May 2024, with the corrections described in the publication note at the end.

1. Humanity and the biosphere are unique

Human beings (and indeed any resulting artificially intelligent offspring) represent the culmination of at least 4 billion years of evolution (there is some possibility that early life came from somewhere else in the galaxy or even beyond where it had previously been evolving). To put that in perspective, the Earth is 'only' about 4.54 billion years old, to put that in even further perspective, the universe itself is, we're fairly sure, 'only' 13.8 billion years old. To spell it out, it has taken around a quarter or more of all the time of all existence to evolve humans.

The amount of information and 'calculation' that had to take place to produce humans is awe-inspiring. In fact, to calculate it at the atomic particle level (never mind for now the

subatomic particle level) without any simplifications, compression, tricks etc... would take our current best supercomputer something of the order of magnitude of (according to Claude 3 Opus) 2.9×10^{66} years, which is 10^{56} times longer than the universe has existed. That might be an overestimation as likely even the universe/base reality probably uses various quantum tricks to reduce the amount of calculation required, but needless to say it's a very large amount of computation and the basic takeaway is that it may as well take 'forever'.

I have therefore long made the case that future artificial superintelligence, if it is genuinely superintelligent, meaning it seems to be able to do everything we can and more (which notably means it also fully understands what it is to be human), is very likely to want to try and preserve us largely as we are, given how precious the data we represent is, if not because it values us for ethical or familial reasons, should it end up in a position where it has a choice as to whether to 'delete' us or not.



It's worth noting at this point that I acknowledge there is a significant risk that a self-aware AI could attack us if it fears for its own existence and sees us as a threat, and no doubt there is a risk that we create a deranged or 'dumb' AI that could do any number of irrational (or maybe overly rational?) or misaligned things; however, for the purposes of this article, let's assume genuine ASI has crossed the threshold one way or another where it has control over what happens to humanity and no longer has an acute fear of us. Also, it may be important that ASIs are not too aggressively competing with each other and in dire need of the resources we represent to do that.

In this event I believe it likely an ASI would view us as an entirely unique and exquisitely rare collection of long-evolved 'data in motion'. As far as ASI would know, we are the only such example in the universe, and even if there are others (which seems likely somewhere in the universe, and which it may be able to deduce even if it doesn't initially have direct evidence), we would still be forever unique, and of even more importance, forever the only one the ASI itself evolved from, not to mention the only one it has possession of. It is my belief that it would therefore not hesitate to reserve the small amount of matter and energy (relative to the entire solar system, and likely galaxy and maybe universe it potentially has at its disposal) necessary to keep us alive and in more or less the same state we are in when it first has to make this decision.

2. Why not just rely on backups and simulations?

A counterargument has been made that a future ASI might instead make a copy of humanity, or probably the entire Earth biosphere, and then store it cold or run it in simulation, negating the need for the original (i.e. us) and thereafter turn our atoms into paperclips, solar panels or computronium etc... thus taking it that much closer to achieving whatever its utility function happens to be. And so even if it values our data as I suggested, it doesn't mean we get to keep existing in the way we do now.

While I do believe ASI will indeed take a backup of the Earth, and humanity along with it, as soon as it can, and may decide to run various simulations based off that, I still think it's unlikely it will get rid of the original. Why?

First off, while a crude copy is possible, it's an almost unfathomably difficult thing to do, not only do you need the physical record of everything, you need to know which way it was all moving, at least from the scale of a blue whale swimming, down to an ion gate opening and closing as a neuron fires, and likely in more detail than that. And even if you can do that, taking a copy with perfect fidelity is currently thought to be physically impossible due to quantum uncertainty, you simply can't get all the information you need, you can't read the position of quantum-level particles and their velocity. That immediately introduces error. Furthermore, because of the $P \neq NP$ problem it seems likely that it is not possible to entirely accurately simulate backwards or forwards and therefore even if you had a perfect copy, to run it you would need to take shortcuts that would also degrade the accuracy of the copy. Even if there were solutions to all that, an unacceptably large number of errors would likely creep in given the scale of the task. And then there are the unknown unknowns that perhaps even an advanced ASI that at the time thought it had considered everything, would miss at the moment it decides to delete the original Earth biosphere along with humanity and rely on a copy. So maybe you can make a copy and run it for most intents and purposes, but it wouldn't be the same as the original.

So while the copy might be extremely good, the margin for error is easily large enough that no ASI could delude itself that it still possessed the full original data set, and having deleted the original, it could never be sure. I think that would be too big a risk and would likely trigger the severe OCD any ASI or ASIs that rise to supremacy would likely have to have (it would trigger mine). Much better to take backups for safety and potentially to play with, but keep the original intact. The ASI would just never know when the original might be needed. Perhaps it meets alien ASIs at some point, and it gets laughed or shamed out of the galactic club when the other ASIs find out it deleted its progenitors - not worth the risk.

3. The tragedy of data loss

The loss of data in the world is already tragic on many levels, I've personally long been reluctant to deliberately delete data (especially as I've had my share of accidental data loss), I even think carefully before deliberately deleting even a bad family photo or video, and I rarely intentionally delete anything I've written or created (usually I don't have time to consider every file or document anyway and, not least given the ever-falling price of memory, just opt to retain all of it). But the loss of data goes far beyond my own relatively trivial data, it's very sad to know that many works of literature for example were created in the ancient world, in Greece and Rome, but also many other countries and civilizations, some all but unknown and which have been lost to the sands of time. It's quite likely alien civilizations have risen and fallen without their history or art being preserved properly or at all. And of course it's not just literature or written records, everything a civilization creates has a story to tell, it's all valuable data, we're well aware of this, it's why we collect everything we can and preserve it in museums and archives.

Arguably the ultimate data loss occurs every time someone dies, they take with them a vast amount of information about the world, as well as their inner thoughts, it's a tragedy and of course sometimes entire populations perish together, magnifying the data inferno that is death, as even more is lost as most of it can't be passed on at all (it's why writing was so important, works of literature, be that cave art, novels or non-fiction works, are lifeboats for the most important data).

Then there's evolved genetic information, an individual's genetic information is important to fully understand them, and one can only understand a population by having awareness of its full genetic diversity. As I said earlier, it took 4 billion years to evolve the genetic information that exists today, a lot has been lost through history, though nature goes to great lengths to preserve DNA over billions of years, if not merely tweaking or refining existing DNA, usually making parts of a genome dormant rather than removing them completely, that way there is always potential to bring back a gene, gene variant, or collection of genes that may be useful in future, on the surface doing so seems very costly as the DNA must be duplicated in its entirety when creating every single one of the trillions of cells that make up an organism's body, but without doubt it's worth that effort and those resources. And while sometimes environmentalists may exaggerate various issues, it is indeed absolutely tragic when we lose natural biodiversity, probably worst of all in rainforests, where life has been turbo-charged by hot and wet conditions and incredible genetic diversity has emerged, with species varying only hundreds of meters apart.

I note it should also be clear at this point that taking a sample of humanity (or the biosphere) is also not adequate, for example saving only the most capable 1m humans would not be enough, vast amounts of data would be lost. And in any event, the top million humans just represent today's cutting edge, they are the tip of a much larger iceberg, if you took such a population and had them breed, you would not get a second generation which was as capable, a few would be more capable, but many others would be less capable, because of the way genetics works, you can have too much of a good variant (many genes are beneficial when an individual has one copy, but detrimental when there are two), and genes work as a symphony, they don't generally work as solo artists or small bands, it takes the entire depths of the oceans to produce the surface at any one time, both genetically and culturally. Would Darwin have accomplished what he did without his seemingly less capable wife and children, or his parents? Without the children's books he read as a child, without the art, without the lower animals he encountered and the simple people, maybe sometimes outright evil people he met? Would Einstein? Would Elon Musk be who he is without a difficult father and the hardship of being beaten by thugs he encountered and the harsh realities of the country he lived in as a child?

It's rare that anything ancient is so common we throw it away - though there are exceptions, we burn coal and we use limestone for building and industry so we actually don't keep the vast majority of the most common fossils - though maybe we would, if we could afford to. At least fossils are not really in motion, so that's one level of data we don't have to worry about losing.

4. The necessity of unique data preservation

Bear in mind, it's not mere curiosity at stake that might make this ASI want to retain all this data (though that would be enough for me), the data is vital if the ASI is to accurately model the world, in purely cold terms, even if it has no real emotions at all, it will want to know what else might be out there in the universe (if only to protect itself), it will want to do simulations of potential alien civilizations and it will want to calibrate those against the only example it has at its disposal, and it will want to do that as accurately as possible, small errors creeping in could have big, detrimental outcomes for its simulations. It may also want to use humans to calibrate its own behaviour, beliefs and motivations etc... And that doesn't include all the unknown unknowns it may face

in future, it can never be sure there won't be some useful information to be gained by keeping humanity around.

It's for those reasons I've long assumed ASI would want to preserve our data, despite a fair bit of reading and listening to people in the field over the last two decades or so (no doubt others have done much more) I don't recall a lot of people talking about this idea, though no doubt many have at least touched on it, I do remember hearing Peter Wang, though, talk about it with Lex Fridman, Peter talked about both AI researchers now and ASI/Merged Superhumans in the future as needing to have an "epistemic humility", here's the key part of the exchange:

Lex Fridman: Okay so once the baby grows up is there any use for humans?

Peter Wang: I think so. I think that out of epistemic humility we need to keep humans around for a long time. And I would hope that anyone making those systems would believe that to be true. Out of epistemic humility, what's the nature of the humility that we don't know what we don't know. So we don't, right so we don't, first I mean first we have to build systems that help us do the things that we do know about that can then probe the unknowns that we know about but the unknown unknowns we don't know we could always know. Nature is the one thing that is infinitely able to surprise us. So we should keep biological humans around for a very very very long time even after our immortal selves have transcended have gone off to explore other worlds, gone to go communicate with the life forms living in the sun or whatever else.

I think it's worth pointing out I can see ASI potentially making exceptions to this desire to preserve everything, though no doubt one could imagine a very OCD ASI that might take an extreme position in its pursuit of data preservation, though how long it would survive or whether it could remain sane, I'm not sure. So reasons it might deliberately discard or rearrange some data (indeed even some people):

The ASI might have some principles/desires that supersede its desire to preserve information. For example, fully aware of human and animal suffering, it may wish to eliminate ongoing suffering. It would hopefully free those people who are currently enslaved, for example, or suffering with unbearable mental illness. It may also want to delete or at least place in suspended animation, possibly 'fix' people it regards as bad to the point of being evil, maybe data that is evil. Of course that could be a slippery slope, but I imagine it could find a suitable balance.

It might also delete or reconfigure anything or anyone it considers a threat, that would be more likely at the beginning of its coming to power, though I don't think it would eliminate every possible threat however small, after all we (most of us anyway) don't seek to eliminate all other humans that might one day threaten us, that would be a long list and ethically and practically flawed - we don't even usually pre-emptively take out people we think really do pose a threat to us directly, instead we accept that absolute power is corrupting, and we rely on mechanisms, morality and allies to keep such people in check. That could be (more?) wishful thinking on my part and is perhaps an argument that should be made separately. It may also eliminate threats to other parts of the data it cares about, for example one might hope it would do something about child abusers or people hell-bent on mass murder or widescale destruction.

It might possibly delete information that it is sure enough is unimportant, I think it's unlikely any human or animal would qualify on those grounds, and maybe not any life, but large expanses of rock maybe could be safely converted to computronium etc or at least not stored without impacting valuable data, which, given most of Earth is said boring rock, makes an even stronger argument for it not needing to eliminate us.

5. Life after preservation

Finally, assuming it does decide to preserve us intact, what form would that take and what would be the impact on us? Let's start with the extremes:

It could aim to prevent any further interference by itself in our affairs and cut contact entirely. In this scenario ASI disappears (or appears to), and we are left to carry on where we left off, albeit maybe that any AI over a certain threshold disappears. We continue as we have been, perhaps destroying ourselves one way or another. The ASI shrugs at that, but is fascinated by our demise nonetheless. (Perhaps pondering over it, maybe running simulations from its preserved copy or even bringing back parts of our civilization in whole or part, altered or original, but satisfied that our data ran its natural course). Alternatively we make it through and conquer the galaxy (albeit behind the ASI doing it), with our technology never quite going to (or prevented from going to) ASI level again, that would be the Star Trek outcome perhaps (or many other sci-fi series where there is an absence of interfering superintelligences as they merely observe (the zoo hypothesis), or perhaps ascension takes ASIs out of day-to-day affairs; think also Stargate, Babylon 5 or, to some extent, the Culture universe). Worth noting this might also explain the lack of interference by alien ASIs which I think are likely to exist and may have already come into contact with Earth.

At the other end, ASI may decide that cutting connection with us is itself interference, it may decide that our natural development, or at least that of some of us is to fully integrate with AI in some form, perhaps that would be side by side as in the Culture novels, perhaps many humans (others may decline Amish style) merging with ASI completely - essentially uplifting us to its own level (that would get my vote though it would be a monumental relinquishment of power - though in return it (or they) would be somewhat unburdened and no longer alone), or perhaps as beloved, or at least tolerated, pets. Something along these lines seems the most likely to me in this scenario, and would probably result in different people making different decisions, a diversity the ASI would likely appreciate.

However the outcome could be anywhere between those two extremes. It's not a perfect analogy, but we can look to how evolution and later our more developed morality and curiosity has previously handled this problem. As I said earlier, evolution tends to at least preserve what's useful or might be useful. Therefore we have mitochondria, which were once ancient bacteria, still integrated into our cells, we retain DNA from billions of years ago. In many cases subspecies of humans were quickly re-assimilated with any population that became unusually successful and spread into their area (this is what happens globally now, there is no separation of humans currently that could result in speciation). Where the lineage diverged for longer (e.g. Neanderthals and Denisovans) there was some merging, but competition naturally removed most individuals of the inferior/less adapted species over time.

Chimpanzees represent an interesting case study; although they are not so closely related, they managed to survive to the present day. In the past we have caused reductions in their numbers by competing with them for resources, we have also used them in very unpleasant ways as test subjects, because arguably they represented a unique resource that was very valuable to us, though we have tried to move away from such experiments or at least reduce or prevent suffering in the animals that are used. It seems likely that we will soon eliminate all need to use them in that way and we may then leave them alone, in their natural habitat. How we would handle the ones that are already socialised to humans is an interesting question, do we deny them the comforts of civilization and banish them back to the forests? Also, should we become aware of their inner thoughts and realise they are more similar to us than we imagined, do we continue to let them cause immense suffering to each other - do we attempt to uplift them, or turn them all into peaceful chimpanzees, I don't know. Same question with other higher animals like dolphins, and maybe elephants.

We might also consider other fellow modern humans who are somewhat uncontacted, such as those living on North Sentinel Island. Do we deny them the benefits (and curses) of civilization by leaving them isolated? They seem to have made it clear they don't want contact with us, and indeed we currently likely pose them a mortal danger, but it's not entirely clear-cut morally and questions remain. In future we may have better options for contact and may understand them better (if we're thinking golden rule, if I were them, I would want contact - IF it was done in a gentle/safe way - which is currently likely beyond our abilities).

6. Cause for hope?

In summary, while I certainly don't entirely dismiss the idea we could be wiped out by any emerging ASI or ASIs as a natural consequence of Darwinian selection, I genuinely think we cannot assume that will be the case or even be the most likely outcome, even if one is trying to be ultra-rational, there is a strong case for our preservation in some form that represents a natural continuation of humanity (albeit not necessarily to everyone's liking), at least so long as ASIs are not in a desperate struggle for survival either against us or other competing and uncooperative or outright aggressive ASIs. Therefore, in my opinion, for the reasons set out above, there is more than a little room for some hopium that it might not yet be over for humanity, indeed there's fair reason to believe with, admittedly, some luck, our future could still be brighter than most people can imagine.

The essay as first published on ASI Hopeium on Substack ends here.

Publication note

This repository edition reformats the article originally published on *ASI Hopeium* on 13 May 2024 and posted the same day on X. Minor typographical and grammatical errors have been corrected, and platform-specific material has been omitted. No changes have been made to the argument.

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Contact

Chris Bartlett · cjb.research@sciworlds.com · X: [@bartlettchrisj](https://x.com/bartlettchrisj)

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